ON-BOARD DIAGNOSTIC [ENTERTAINMENT SYSTEM (WITH CENTER DISPLAY)]

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Note
- “iPod” is a registered trademark of Apple Inc.

Outline
- The entertainment system has an on-board diagnostic function and diagnostic assist function to facilitate system diagnosis.

On-board diagnostic function
- The on-board diagnostic function consists of the following functions:
  - — A malfunction detection function, which detects overall malfunctions in the entertainment system-related parts (with center display)
  - — A memory function, which stores detected DTCs
  - — A display function, which indicates system malfunctions via a DTC display.

Malfunction detection function
- Detects malfunctions in the entertainment system.

Memory function
- If a malfunction is detected in the entertainment system, the connectivity master unit (CMU) stores a DTC.

Display Function
Mazda Modular Diagnostic System (M-MDS) not used
- To display DTCs on the center display that have been recorded in the memory function, activate the on-board diagnostic function by operating the center display or commander switch. Refer to the [DTC INSPECTION [CONNECTIVITY MASTER UNIT]] in the workshop manual for the activation procedure of the on-board diagnostic function.

Mazda Modular Diagnostic System (M-MDS) used
- When DTC inspection is performed using the Mazda modular diagnostic system (M-MDS), the audio system malfunction diagnosis begins. Refer to the [DIAGNOSTIC ASSIST FUNCTION [CONNECTIVITY MASTER UNIT]] in the workshop manual for the DTC troubleshooting procedure.

DTC table

×: Applicable
—: Not applicable

DTC No.	Diagnostic assist code*1	Warning/ indicator lights	Description	Fail-safe	Drive cycle	Self test type*2	Memory function
B100E:87*3	03/04	—	Communication error with DVD/ CD player	—	—	C	×
B1087:87	03/04	—	Communication error with commander switch	—	—	C	×
B108E:04	03/04	—	Center display internal malfunction	—	—	C	×
B108E:06	03/04	—	HDCP authorization error	—	—	C	×
B108E:1C	03/04	—	Low power supply voltage input to center display	×	—	C	×
B108E:87	03/04	—	Center display circuit malfunction	—	—	C	×
B116A:11	03/04	—	Microphone circuit malfunction	—	—	C	×
B116A:12	03/04	—	Microphone circuit malfunction	—	—	C	×
B116A:92	03/04	—	Microphone circuit malfunction	—	—	C	×
B119F:11	03/04	—	GPS antenna circuit malfunction	—	—	C	×
B119F:13	03/04	—	GPS antenna circuit malfunction	—	—	C	×
B119F:1D	03/04	—	GPS antenna circuit malfunction	—	—	C	×
B119F:92	03/04	—	GPS antenna circuit malfunction	—	—	C	×
B11BA:11	03/04	—	Steering switch circuit malfunction	—	—	C	×
B11BA:15	03/04	—	Steering switch circuit malfunction	—	—	C	×

DTC No.	Diagnostic assist code*1	Warning/ indicator lights	Description	Fail-safe	Drive cycle	Self test type*2	Memory function
B1201:11	03/04	—	Steering switch circuit malfunction	—	—	C	×
B1201:15	03/04	—	Steering switch circuit malfunction	—	—	C	×
B1238:01	03/05	—	Speaker circuit malfunction	—	—	C	×
B1238:16	03/05	—	Tuner and amp unit (TAU) low power supply voltage input	—	—	C	×
B1238:87	03/04	—	RDS traffic information transfer circuit malfunction	—	—	C	×
B1252:09	03/04	—	Auxiliary jack/USB port/SD card slot hub circuit malfunction	—	—	C	×
B1252:1D	03/04	—	Auxiliary jack/USB port/SD card slot hub circuit malfunction	—	—	C	×
B1252:87	03/04	—	Communication error with auxiliary jack/USB port/SD card slot hub	—	—	C	×
B12BD:11*3	03/04	—	Rear mount camera circuit malfunction	—	—	C	×
B12BD:87*4	03/04	—	Rear mount camera transmission error	—	—	C	×
B1A89:13 *5	03/05	—	DAB antenna feeder malfunction	—	—	C	×
B1A89:42 *5	03/05	—	Tuner and amp unit internal malfunction	—	—	C	×
B1D19:00 *3	03/06	—	Region code error or parental lock function error	—	—	C	×
B1D19:64 *6	03/06	—	Incompatible disc inserted	×	—	C	×
B1D19:79 *6	03/06	—	CD player or DVD/CD player disc read error	×	—	C	×
B1D19:98 *6	03/06	—	Abnormal CD player or DVD/CD player temperature	×	—	C	×
U0001:88	03/04	—	Module communication error (HS-CAN)	—	—	C	×
U0028:88	03/04	—	Unit communication error (LOCAL-CAN)	—	—	C	×
U0100:00	03/04	—	Communication error with PCM	—	—	C	×
U0101:00 *7	03/04	—	Communication error with TCM	—	—	C	×
U0121:00	03/04	—	Communication error with ABS HU/CM or DSC HU/CM	—	—	C	×
U0131:00	03/04	—	Communication error with EPS control module	—	—	C	×
U0140:00	03/04	—	Communication error with BCM	—	—	C	×
U0151:00	03/04	—	Communication error with SAS control module	—	—	C	×
U0155:00	03/04	—	Communication error with instrument cluster	—	—	C	×
U0182:00 *8	03/04	—	Communication error with auto leveling control module	—	—	C	×
U0184:00	03/04	—	Communication error with tuner and amp unit (TAU)	—	—	C	×
U0187:00 *6	03/04	—	Communication error with CD player or DVD/CD player	—	—	C	×
U0214:00	03/04	—	Communication error with start stop unit	—	—	C	×
U0235:00 *9	03/04	—	Communication error with laser sensor	—	—	C	×

DTC No.	Diagnostic assist code*1	Warning/ indicator lights	Description	Fail-safe	Drive cycle	Self test type*2	Memory function
U023A:00*10	03/04	—	Communication error with forward sensing camera (FSC)	—	—	C	×
U2300:54	03/04	—	Configuration data not received from instrument cluster	—	—	C	×
U2300:55	03/04	—	Instrument cluster configuration not implemented	—	—	C	×
U3000:42	03/04	—	Connectivity master unit (CMU) internal malfunction (poor data read-outs)	—	—	C	×
U3003:16	03/04	—	Low ACC power supply voltage input to connectivity master unit (CMU)	—	—	C	×

*1 : The double-digit number indicates the diagnostic assist code with which DTCs can be detected during the DTC inspection using the diagnostic assist function.
*2 : C: CMDTC self test, D:ODDTC self test
*3 : With DVD/CD player
*4 : With rear mount camera
*5 : With DAB antenna amplifier
*6 : With CD player or DVD/CD player
*7 : ATX
*8 : With headlight auto levelling system
*9 : With smart city brake support (SCBS)
*10 : With forward sensing camera (FSC)

Detection condition for the applicable DTC

DTC No.	Description	Fail-safe
B100E:87	Communication error with DVD/CD player	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a signal from the DVD/CD player.
B1087:87	Communication error with commander switch	• The connectivity master unit (CMU) cannot receive a LIN signal from the commander switch for **5 s or more** with the ignition switched ACC or ON (engine off or on).
B108E:04	Center display internal malfunction	• The connectivity master unit (CMU) detected a center display internal malfunction for **5 s or more** with the ignition switched ON (engine off or on).
B108E:06	• The connectivity master unit (CMU) detected an HDCP authorization error for **5 s or more** with the ignition switched ON (engine off or on).	• The connectivity master unit (CMU) detected an HDCP authorization error for **5 s or more** with the ignition switched ON (engine off or on).
B108E:1C	Low power supply voltage input to center display	• The connectivity master unit (CMU) detected any of the following conditions for **5 s or more** with the ignition switched ON (engine off or on). — TFT LCD driver voltage **3.8 V or less** — LED drive voltage **14.0 V or less** — Touch panel IC voltage **3.9 V or less**
B108E:87	Center display circuit malfunction	• The connectivity master unit (CMU) cannot receive a signal from the center display for **5 s or more** with the ignition switched ON (engine off or on).
B116A:11	Microphone circuit malfunction	• The connectivity master unit (CMU) detected a short to ground in the microphone circuit for **5 s or more** with the ignition switched ON (engine off or on).
B116A:12	Microphone circuit malfunction	• The connectivity master unit (CMU) detected a short to power supply in the microphone circuit for **5 s or more** with the ignition switched ON (engine off or on).
B116A:92	Microphone circuit malfunction	• The connectivity master unit (CMU) detected microphone circuit voltage of **less than 7.7 V** or **more than 8.6 V** with the ignition switched ON (engine off or on).

DTC No.	Description	Fail-safe
B119F:11	GPS antenna circuit malfunction	• Connectivity master unit (CMU) detects GPS antenna circuit voltage of **4.45 V or less** with the ignition switched ON (engine off or on).
B119F:13	GPS antenna circuit malfunction	• The connectivity master unit (CMU) detects an open circuit in the GPS antenna circuit with the ignition switched ON (engine off or on).
B119F:1D	GPS antenna circuit malfunction	• The connectivity master unit (CMU) detects a current value in the GPS antenna circuit of (**31 mA or more**) with the ignition switched ON (engine off or on).
B119F:92	GPS antenna circuit malfunction	• The connectivity master unit (CMU) cannot receive signals from the GPS antenna for **5 s or more** with the ignition switched ON (engine off or on).
B11BA:11	Steering switch circuit malfunction	• The connectivity master unit (CMU) detected a short to ground in the steering switch circuit for **5 s or more** with the ignition switched ON (engine off or on).
B11BA:15	Steering switch circuit malfunction	• The connectivity master unit (CMU) detected an open circuit or a short to the power supply in the steering switch circuit for **5 s or more** with the ignition switched ON (engine off or on).
B1201:11	Steering switch circuit malfunction	• The connectivity master unit (CMU) detected a short to ground in the steering switch circuit for **5 s or more** with the ignition switched ON (engine off or on).
B1201:15	Steering switch circuit malfunction	• The connectivity master unit (CMU) detected an open circuit or a short to the power supply in the steering switch circuit for **5 s or more** with the ignition switched ON (engine off or on).
B1238:01	Speaker circuit malfunction	• The connectivity master unit (CMU) detected a short to ground or short to power supply in the speaker circuit with the ignition switched ON (engine off or on).
B1238:16	Tuner and amp unit (TAU) low power supply voltage input	• Connectivity master unit (CMU) detects power supply circuit voltage for tuner and amp unit (TAU) of **less than 10 V** for **10 s or more** with the ignition switched ON (engine off or on).
B1238:87	RDS traffic information transfer circuit malfunction	• The connectivity master unit (CMU) detects a communication error in the RDS traffic information transfer circuit for **20 times** with the ignition switched ON (engine off or on).
B1252:09	Auxiliary jack/USB port/SD card slot hub circuit malfunction	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) detects a voltage value for the auxiliary jack/USB port/SD card slot hub circuit of **within 4.8 V** or **exceeding 5.2 V**.
B1252:1D	Auxiliary jack/USB port/SD card slot hub circuit malfunction	• The connectivity master unit (CMU) detects over-current (**550 mA or more**) in auxiliary jack/USB port/SD card slot hub circuit with the ignition switched ON (engine off or on).
B1252:87	Communication error with auxiliary jack/USB port/SD card slot hub	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a signal from the auxiliary jack/USB port/SD card slot hub.
B12BD:11	Rear mount camera circuit malfunction	• The connectivity master unit (CMU) detects a voltage value for the rear mount camera circuit of **within 5.8 V** or **exceeding 6.2 V for 5 s or more** with the ignition switched ON (engine off or on).
B12BD:87	Rear mount camera transmission error	• The connectivity master unit (CMU) cannot receive a signal from the rear mount camera for **5 s or more** with the ignition switched ON (engine off or on).
B1A89:13	DAB antenna feeder malfunction	• The connectivity master unit detects an open circuit in DAB antenna feeder with the ignition switched ON (engine off or on).
B1A89:42	Tuner and amp unit internal malfunction	• The connectivity master unit (CMU) detects a region code error or parental lock function error with the ignition switched ON (engine off or on).
B1D19:00	Region code error or parental lock function error	• The connectivity master unit (CMU) detects a region code error or parental lock function error with the ignition switched ON (engine off or on).
B1D19:64	Incompatible disc inserted	• With the ignition switched ON (engine on or off), the connectivity master unit (CMU) detects a disc having a data format that is incompatible with the CD player or DVD/CD player.
B1D19:79	CD player or DVD/CD player disc read error	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) detects a disc read error.
B1D19:98	Abnormal CD player or DVD/CD player temperature	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) detects an abnormally high temperature condition in the CD player or DVD/CD player.

DTC No.	Description	Fail-safe
U0001:88	Module communication error (HS-CAN)	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot detect a CAN bus communication line (HS-CAN) signal for **1 s or more**. • With the ignition switched ON (engine off or on), the connectivity master unit (CMU) detects malfunction in the CAN bus communication line (HS-CAN) **10 times** continuously.
U0028:88	Unit communication error (LOCAL-CAN)	• The connectivity master unit (CMU) detects CAN bus communication line (LOCAL-CAN) malfunction **10 times** continuously.
U0100:00	Communication error with PCM	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the PCM for **5 s or more**.
U0101:00	Communication error with TCM	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the TCM for **5 s or more**.
U0121:00	Communication error with ABS HU/CM or DSC HU/CM	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the ABS HU/CM or DSC HU/CM for **5 s or more**.
U0131:00	Communication error with EPS control module	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the EPS control module for **5 s or more**.
U0140:00	Communication error with BCM	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the BCM for **5 s or more**.
U0151:00	Communication error with SAS control module	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the SAS control module for **5 s or more**.
U0155:00	Communication error with instrument cluster	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the instrument cluster for **5 s or more**.
U0182:00	Communication error with auto leveling control module	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the auto leveling control module for **5 s or more**.
U0184:00	Communication error with tuner and amp unit (TAU)	• The connectivity master unit (CMU) cannot receive a CAN signal from the tuner and amp unit (TAU) for **5 s or more**.
U0187:00	Communication error with CD player or DVD/CD player	• The connectivity master unit (CMU) cannot receive a CAN signal from the CD player or DVD/CD player for **5 s or more**.
U0214:00	Communication error with start stop unit	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the start stop unit for **5 s or more**.
U0235:00	Communication error with laser sensor	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the laser sensor for **5 s or more**.
U023A:00	Communication error with forward sensing camera (FSC)	• With the ignition switched ON (engine off or on), the connectivity master unit (CMU) cannot receive a CAN signal from the forward sensing camera (FSC) for **5 s or more**.
U2300:54	Configuration data not received from instrument cluster	• The connectivity master unit (CMU) cannot receive the configuration data from the instrument cluster for **30 s or more** with the ignition switched ON (engine off or on).
U2300:55	Instrument cluster configuration not implemented	• The connectivity master unit (CMU) detected any of the following conditions with the ignition switched to ACC or ON (engine off or on). — Instrument cluster configuration not read — Configuration data cannot be written to connectivity master unit (CMU) memory.
U3000:42	Connectivity master unit (CMU) internal malfunction (poor data read-outs)	• The connectivity master unit (CMU) cannot read data from the NVRAM in the connectivity master unit (CMU) with the ignition switched to ACC or ON (engine off or on).
U3003:16	Low ACC power supply voltage input to connectivity master unit (CMU)	• The connectivity master unit (CMU) detects ACC power supply circuit voltage of **8.0 ± 0.8 V or less** for **10 s or more** with the ignition switched to ACC or ON (engine off or on).

Diagnostic Assist Function

- The diagnostic assist function is a function for forcibly operating entertainment system related parts to verify their connection condition and operation status. In addition, the specifications of the entertainment system related parts are displayed in the center display where they can be verified. Refer to the [DIAGNOSTIC ASSIST FUNCTION [CONNECTIVITY MASTER UNIT]] in the workshop manual for the diagnostic assist function activation procedure.

Diagnostic assist code list

No.	Center display (diagnostic content)	Function	Content
01	CD Test	CD player or DVD/CD player inspection	The CD player or DVD/CD player inspection results are displayed in the center display.
02	Clear DTC	DTC clearing	Clears the DTCs. **Note** • DTC clearing is performed in the following order, and a message is displayed each time to indicate the DTCs are cleared. — Connectivity master unit (CMU) — Tuner and amp unit (TAU) — CD player or DVD/CD player
03	Read AVC DTC	Inspection of all DTCs related to module	Displays the DTCs of the entertainment system related parts. **Note** • Detection of DTCs is limited if the ignition is in the ACC position. Refer to the workshop manual for details.
04	Read CMU DTC	DTC inspection for connectivity master unit (CMU)	The connectivity master unit (CMU) DTCs are displayed in the center display.
05	Read TAU DTC	DTC inspection for tuner and amp unit (TAU)	The tuner and amp unit (TAU) DTCs are displayed in the center display.
06	Read CD DTC	DTC inspection for CD player or DVD/CD player	The CD player or DVD/CD player DTCs are displayed in the center display.
07	Read DTV DTC	Does not operate due to unavailability of applicable DTC.	
08	Software Reset AVC	Do not operate because item is for manufacturer use.	
09	Software Version Display (CD)	Software version verification	Displays the CD player or DVD/CD player software version.
10	Part Number Readout CD/DVD	Part number verification	Displays the CD player or DVD/CD player part number (first four digits)
21	DVD Test Mode Enable	Do not operate because item is for manufacturer use.	
22	DVD Test Mode Disable	Do not operate because item is for manufacturer use.	
24	Vehicle Data Transfer	Do not operate because item is for manufacturer use.	
25	Antenna Continuity Status	Do not operate because item is for manufacturer use.	
26	DTV Test Mode ON	Do not operate because item is for manufacturer use.	
27	DTV Test Mode OFF	Do not operate because item is for manufacturer use.	
28	HD Certification AM (Main Tuner)	Do not operate because item is for manufacturer use.	
53	Software Version Display (CMU)	Software version verification	Displays the connectivity master unit (CMU) (overall entertainment system (with center display)) software version.
57	Software Version Display (VIP)	Software version verification	Displays the connectivity master unit (CMU) (overall communication with vehicle) software version.
58	Vehicle Info	Configuration information verification	Displays the following configuration information in the center display. — Destination — CMU Types
59	CMU Serial Number Readout	Individual number verification	Displays the unique number verification of the connectivity master unit (CMU) in the center display.

No.	Center display (diagnostic content)	Function	Content
61	Vehicle Signal (Unit Status)	Vehicle signal verification	Displays the following vehicle signal information in the center display. — Rear mount camera signal input condition — Steering switch signal (volume adjustment switch, cue/automatic band selector switch) input condition — Steering switch signal (talk button, pick-up button, hang-up button) input condition — Commander switch signal input condition — GPS antenna signal input condition — Voice recognition-use microphone signal input condition — Auxiliary jack/USB port/SD card slot hub 1 signal input condition — SD card insertion condition — CD or DVD equipment recognition condition — TV tuner unit equipment recognition condition
65	Commander and Switch Check	Switch inspection	Displays the steering switch or commander switch operation content in the center display. **Note** • If the operation content previously detected is being displayed in the center display, newly detected content is added over it.
68	Software version display (TFT Display)	Software version verification	Displays the center display (TFT LCD screen control) software version.
69	Software version display (Touch Panel)	Software version verification	Displays the center display (touch panel control) software version.
70	Display Check	Display/touch panel inspection	Displays the display/touch panel inspection results in the center display. 1. Unicolor screen in the following colors — White→Black→Red→Green→Blue 2. Horizontal-striped screen in black and the following colors — Red→Green→Blue 3. Vertical-striped screen in black and the following colors — Red→Green→Blue 4. Checkered screen in black and the following colors — Red→Green→Blue 5. Returns to 1. **Note** • The display/touch panel inspection is repeated periodically until the same location on the center display is touched for **10 s or more**.
72	GPS Data	GPS information verification	Displays the following GPS information in the center display. — Longitude (X) (Degree of longitude in GPS information of the current location) — Latitude (Y) (Degree of latitude in GPS information of the current location) — Absolute Time (Z) (Date and time information (GMT)) — Number of Satellites (Number of satellites being received)
84	Maker code Display TAU	Manufacturer verification	Displays the tuner and amp unit (TAU) manufacturer in the center display.
85	Part Number Readout TAU	Part number verification	Displays the tuner and amp unit (TAU) part number (first 4 digits).

No.	Center display (diagnostic content)	Function	Content
86	Radio Test Mode AM (Main)	Do not operate because item is for manufacturer use.	
87	HD Certification FM (Main Tuner)	Do not operate because item is for manufacturer use.	
88	HD Certification FM (Sub Tuner)	Do not operate because item is for manufacturer use.	
89	XM Certification	Do not operate because item is for manufacturer use.	
90	AM Reception Sensitivity Level Test	Radio reception condition inspection	Displays the radio reception condition in the center display.
91	Radio Test Mode FM (Main Tuner)	Do not operate because item is for manufacturer use.	
92	Radio Test Mode FM (Sub Tuner)	Do not operate because item is for manufacturer use.	
93	Software Version Display (TAU)	Software version verification	Displays the tuner and amp unit (TAU) software version.
94	Speaker Cyclic Test (FM)	Speaker Inspection	Outputs sound from the speakers in the following order: 1. Front door speaker (LH) and tweeter (LH) 2. Front door speaker (RH) and tweeter (RH) 3. Rear door speaker (RH) 4. Rear door speaker (LH)
95	Vehicle Info Display (TAU)	Do not operate because item is for manufacturer use.	
96	XM Serial Number	Do not operate because it is displayed, but, not function.	
97	Radio Electric Field Strength (AM)	Do not operate because item is for manufacturer use.	
98	Radio Electric Field Strength (FM)	Do not operate because item is for manufacturer use.	
99	Software Update	Do not operate because item is for manufacturer use.	